

FIN ST1542–ST1631

Noncirculant Robustness, Continuum Minimax,
Continuous Prior and Fiber Identifiability

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

Six research rounds — 90 programmes
Release 11.01 — 31 August 2026

This report extends the conditional FIN Operational/State discriminator beyond circulant rounding errors and finite nuisance grids. It proves noncirculant operator-norm location robustness, brackets the continuous minimax value, constructs a continuous-prior e-process theorem, certifies independent calibration maps and resolves fine-fiber identifiability on a declared range.

Abstract

ST1542–ST1631 extend the Release 11.00 robustness ladder in five directions. First, Duhamel estimates remove the circulant requirement for the base return- gap location. If A and B are self-adjoint positive semidefinite graph Laplacians with $\|B - A\| \leq \epsilon$, then heat and unitary propagators differ by at most $t\epsilon$ in operator norm. Classical return changes by at most $t\epsilon$, quantum return by at most $2t\epsilon$, and their gap by at most $3t\epsilon$. Combining this with the certified outside margin proves that every global maximizer remains in $[0.59, 0.60]$ whenever

$$\epsilon < 9.720652006795929 \times 10^{-8}.$$

The bound does not preserve uniqueness inside that interval and does not cover non-positive heat generators or dephasing-generator sensitivity.

Second, the continuous bidirectional-KL minimax value is certified. A finite LP proposes endpoint weights near $(0.318, 0, 0, 0.682)$. An adaptive outward cover proves that this candidate achieves at least 0.05 over the entire (α, γ) rectangle. A fixed two-point dual nuisance measure proves that no allocation can exceed 0.05370588868619479. Therefore

$$0.05 \leq V_* \leq 0.05370588868619479.$$

The exact optimizer remains open.

Third, for a uniform continuous prior over the nuisance rectangle, the integral of pointwise likelihood-ratio martingales is itself a nonnegative mean-one martingale by Tonelli/Fubini. It yields a valid one-sided anytime test. A positive 7×9 Gauss–Legendre rule supplies an executable 63-component finite mixture which is valid independently of its quadrature accuracy; it is not claimed to equal the exact continuous integral.

Fourth, independent classical and quantum calibration maps with up to five percent uniform preparation contamination and two percent false-positive/ false-negative rates are certified directly rather than by a crude triangle bound. The base return ranges remain disjoint, and a complete 1000-cell composite cover gives

$$\text{RSS} \geq 0.022937735365240054.$$

Finally, fine-fiber coupling q is analysed. Heat return $(1 + e^{-2qt})/2$ globally identifies $q \geq 0$ from one nontrivial time. Quantum return $\cos^2(qt)$ is globally aliased but injective on its first monotone branch. At one time the two channels cross repeatedly. The two-time design $(0.6, 1.0)$ has an interval-certified separation on $q \in [0.1, 2]$:

$$\text{RSS} \geq 0.009270580748993435.$$

At $q = 0$ both returns equal one for all times, giving an exact obstruction.

Protocol 11.01 freezes these claims and their boundaries. No platform, physical calibration, state ontology or natural event record is supplied.

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1 Round I: ST1542–ST1556 — noncirculant Duhamel bounds

Let A and B be positive semidefinite self-adjoint graph Laplacians with zero row sums and

$$\|B - A\| \leq \epsilon.$$

No circulant symmetry or shared eigenvectors are assumed.

For heat semigroups, Duhamel's formula gives

$$e^{-tB} - e^{-tA} = - \int_0^t e^{-(t-s)B} (B - A) e^{-sA} ds.$$

Both semigroups are contractions, hence

$$\|e^{-tB} - e^{-tA}\| \leq t\epsilon.$$

The unitary analogue gives the same bound:

$$\|e^{-itB} - e^{-itA}\| \leq t\epsilon.$$

For the selected return effect,

$$|C_B(t) - C_A(t)| \leq t\epsilon, \quad |Q_B(t) - Q_A(t)| \leq 2t\epsilon,$$

and therefore

$$|D_B(t) - D_A(t)| \leq 3t\epsilon.$$

Let $t_c = 0.5945308$ and let $m = 3.089572412462438 \times 10^{-6}$ be the certified gap between the candidate value and every point outside $[0.59, 0.60]$. For $t \leq 10$,

$$D_B(t_c) - D_B(t) \geq m - 3(t_c + 10)\epsilon.$$

Hence all global maximizers of D_B remain in $[0.59, 0.60]$ if

$$\boxed{\epsilon < 9.720652006795929 \times 10^{-8}.$$

For arbitrary symmetric entrywise errors bounded by δ , the crude $\|E\| \leq 12\delta$ condition gives $\delta < 8.10054338996607 \times 10^{-9}$ as sufficient. At $\epsilon = 10^{-8}$ the exclusion margin remains $2.771736488462438 \times 10^{-6}$.

Theorem 1 (noncirculant location robustness). *Within the declared graph-Laplacian norm ball, every global return-gap maximizer remains in $[0.59, 0.60]$ below the stated threshold.*

The theorem does not guarantee a unique critical point after noncirculant perturbation. It also does not propagate uncertainty through the dephasing Lindbladian.

2 Round II: ST1557–ST1571 — continuous minimax value

For the four scheduled times define

$$d_i(\theta) = \min\{D(C_i \| Q_{i,\theta}), D(Q_{i,\theta} \| C_i)\},$$

and

$$V_* = \sup_{w \in \Delta_3} \inf_{\theta \in [0.9, 1.1] \times [0, 100]} \sum_i w_i d_i(\theta).$$

A 41×401 grid LP produces a two-endpoint candidate. Its exact displayed weights depend slightly on grid resolution, so only the structural conclusion is retained: roughly 32% at $t = 0.3$ and 68% at $t = 2.0$.

For an interval $q \in [q_-, q_+]$, Bernoulli KL is jointly convex and minimized at $q = p$ if the probability intervals overlap, otherwise at the nearest endpoints. This analytic rectangle minimum, combined with outward return intervals, permits adaptive subdivision of the continuous nuisance rectangle. Starting from 1000 cells, 380 splits produce 1380 accepted boxes and prove the candidate value is at least 0.05.

For the converse, take nuisance points (1.1, 16) and (1.1, 16.25) with weights 0.7622353358 and 0.2377646642. For any time allocation, its worst-case value cannot exceed the weighted average over these two points, whose largest time coordinate is 0.053705888685...; a rounding safety payment yields the upper bound 0.05370588868619479.

Theorem 2 (continuous minimax value bracket). *For the frozen decimal model, declared times, nuisance rectangle and bidirectional-KL metric,*

$$0.05 \leq V_* \leq 0.05370588868619479.$$

The theorem proves near-optimality in value, not exact weights or a unique active nuisance measure.

3 Round III: ST1572–ST1586 — continuous-prior e-process

Let π be the uniform probability measure on $[0.9, 1.1] \times [0, 100]$. For every nuisance point θ , define

$$L_{\theta,n} = \prod_{i=1}^n \frac{q_{\theta}(X_i|t_i)}{p_C(X_i|t_i)}.$$

Under the classical null, $L_{\theta,n}$ is a nonnegative mean-one martingale. Tonelli/Fubini therefore gives

$$E_n^{\text{cont}} = \int L_{\theta,n} \pi(d\theta), \quad \mathbb{E}_C E_n^{\text{cont}} = 1.$$

Theorem 3 (continuous-prior anytime rejection).

$$\Pr_C \left(\sup_n E_n^{\text{cont}} \geq 100 \right) \leq 0.01.$$

The executable approximation uses positive Gauss–Legendre weights with seven clock nodes and nine dephasing nodes. Their 63-component convex sum is itself a valid finite-mixture e-process regardless of quadrature accuracy. It is not the exact integral. A synthetic off-node alternative $(\alpha, \gamma) = (1.03, 7.3)$ crossed after 243 events; this is a power fixture, not evidence or a uniform stopping guarantee.

Both the exact prior integral and the finite quadrature mixture are one-sided tools for rejecting C . They do not uniformly accept C against all quantum nuisance values.

4 Round IV: ST1587–ST1601 — independent calibration

Each hypothesis receives an independent affine map

$$T_{\epsilon,a,b}(p) = a + (1 - a - b) [(1 - \epsilon)p + \epsilon/12],$$

with

$$0 \leq \epsilon \leq 0.05, qquad 0 \leq a, b \leq 0.02.$$

Because the map is multiaffine in (ϵ, a, b) and monotone in p , extrema occur at the eight calibration vertices.

At $t = 0.6$, the independently calibrated ranges are

$$C_{\text{obs}} \in [0.3879880701, 0.4241102492],$$

$$Q_{\text{obs}} \in [0.7708322380, 0.8271041102].$$

They remain separated by at least 0.3467219888.

The same vertex range construction is applied to every cell of the complete 1000-cell (α, γ) cover. It yields

$$\text{RSS}_{\text{independent}} \geq 0.022937735365240054,$$

so at least one calibrated time differs by 0.07572604466965123.

This upgrades the earlier 5% + 2% status. The old triangle bound failed to certify the box because it ignored the affine geometry; that failure was not a counterexample. The present theorem is still limited to binary affine calibration. Vertex-dependent detector matrices remain open.

Independent uniform click efficiencies in $[0.8, 1]$ do not alter click-conditioned ranges, but attempts and no-clicks remain mandatory.

5 Round V: ST1602–ST1616 — fine-fiber q

For initial fiber state zero,

$$H_q(t) = \frac{1 + e^{-2qt}}{2}, \quad R_q(t) = \cos^2(qt).$$

For every $t > 0$, H_q is strictly decreasing on $q \geq 0$ and

$$q = -\frac{\log(2H - 1)}{2t}.$$

Thus classical q is globally identifiable from one nontrivial return.

Quantum return has aliases

$$q' = \frac{k\pi}{t} \pm q.$$

It is injective only on a specified monotone branch, for example

$$0 \leq q \leq \frac{\pi}{2t}, \quad q \text{quad} q = \frac{\arccos \sqrt{R}}{t}.$$

At $q = 0.37$, $t = 0.6$,

$$H = 0.8207327104, \quad q \text{quad} R = 0.9515203359, \quad q \text{quad} R - H = 0.1307876255.$$

The ideal Bernoulli Fisher informations are approximately 1.006807 for heat and $4t^2 = 1.44$ for the quantum branch.

One time does not always distinguish the channels: at $t = 0.6$ their positive crossings below 10 occur near 1.07725, 3.93441, 6.54466 and 9.16299. Moreover,

$$H_0(t) = R_0(t) = 1$$

for every time, so $q = 0$ is an exact no-go.

For $q \in [0.1, 2]$, a 1900-box outward interval cover at times (0.6, 1.0) proves

$$(R_q(0.6) - H_q(0.6))^2 + (R_q(1) - H_q(1))^2 \geq 0.009270580748993435.$$

At least one residual is at least 0.06808296684558274.

The result is conditional on the q range, clock and fiber-resolving instrument. It does not derive a physical q .

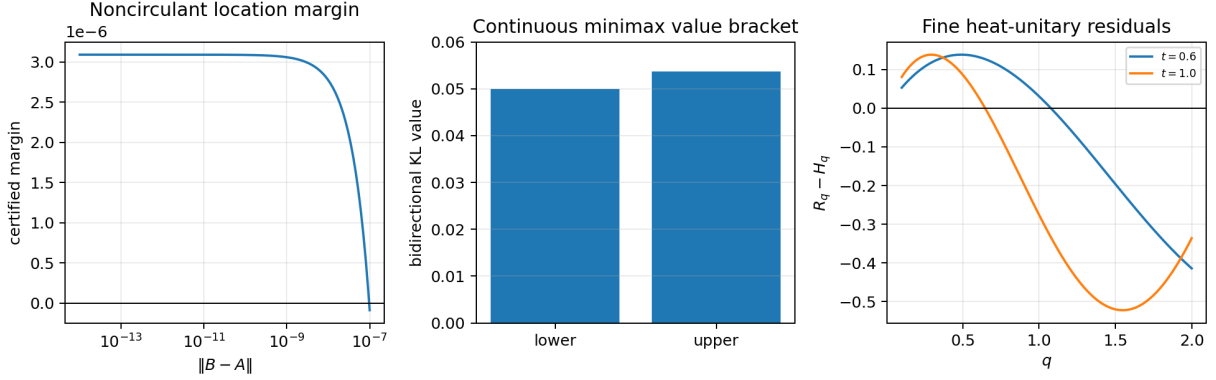


Figure 1: Left: noncirculant location margin versus operator perturbation. Centre: certified continuous minimax value bracket. Right: the two fine-fiber heat-unitary residuals over the certified q range.

6 Round VI: ST1617–ST1631 — synthesis

Result	Status	Boundary
noncirculant maximum location	[Proven bound]	PSD graph Laplacians, norm threshold
continuous minimax value	[Certified bracket]	frozen metric/model
exact continuous-prior e-process	[Proven]	one-sided classical-null rejection
63-node quadrature e-process	[Executable/proven finite]	not exact integral
independent calibration RSS	[Interval-proven]	binary affine maps
classical q inversion	[Proven]	known heat channel
quantum q branch inversion	[Proven conditional]	specified alias-free range
two-time fine channel separation	[Interval-proven]	$q \in [0.1, 2]$
platform/evidence	[Absent]	no external calibration/events

Protocol `FIN_ST1617_ST1631_Protocol_11_01.json` hashes Release 11.00, the continuous-prior quadrature and the independent calibration polytope, and records the noncirculant threshold, minimax bracket and fine-fiber range.

The robustness hierarchy now includes decimal, circulant, noncirculant, continuous-nuisance, independent-calibration and exact-refinement/fine-fiber layers. It still describes one conditional OA , not strict FIN physics.

The experiment tests internal channel/fiber dynamics. It does not test literal nonexistence of the whole nadsoliton.

7 Recommended next programmes

1. derive Duhamel bounds for the full dephasing/Lindblad generator;
2. propagate symbolic $(\omega, \phi, \beta, \eta)$ interval uncertainty;
3. isolate exact minimax weights through KKT/dual equations;
4. construct uniformly valid composite acceptance or confidence sets;
5. certify a vertex-dependent detector matrix polytope;

6. jointly identify fiber q and clock scale;
7. impose multi-level q consistency;
8. optimize the continuous prior for power while preserving validity;
9. derive one platform-specific transfer function;
10. freeze an external calibration form;
11. obtain independent audit;
12. collect raw events only after every preceding freeze.

Stop if location robustness is called uniqueness, a value bracket is called an exact optimizer, quadrature is called exact integration, an affine detector is called an arbitrary matrix, the q theorem is applied at $q = 0$ or outside its range, or synthetic power is called evidence.

8 Final conclusion

The conditional FIN dual-dynamics protocol now has a noncirculant operator-norm location theorem, a certified continuous minimax value, an exact continuous- prior anytime theorem, a positive independent-calibration certificate and an interval-certified fine-fiber test. The remaining obstacles are no longer ordinary finite numerical ambiguity; they are full generator uncertainty, general detector matrices, exact composite decisions and physical realization.

A Six-round ledger

Range	Object	Verdict
ST1542–ST1556	noncirculant perturbation	global location bound
ST1557–ST1571	continuous minimax	value bracket [0.05, 0.053706]
ST1572–ST1586	continuous-prior e-process	exact theorem, finite quadrature
ST1587–ST1601	independent calibration	positive base and composite separation
ST1602–ST1616	fine-fiber q	inversion/aliasing/two-time certificate
ST1617–ST1631	synthesis	six-layer conditional robustness hierarchy

B Reproducibility

The authoritative result sets run from `FIN_ST1542_ST1556_Results.json` through `FIN_ST1617_ST1631_Results.json`. Each contains fifteen programmes and packet SHA-256 references. The combined test `test_fin_st1542_st1631.py` verifies all ninety packets, perturbation thresholds, continuous bracket, prior grid, independent calibration, fiber certificate, protocol, figure and nonpromotion gates.

C Nonclaims

No noncirculant dephasing uncertainty theorem, exact minimax weights, symmetric composite decision, arbitrary detector matrix certificate, physical q source, state ontology, platform, raw event, independent custody, legacy bridge, selector, Standard Model, gravity, L_{total} or Theory of Everything is established. All future FIN PDFs remain English.

References

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